

## Lab 2: A price you can take to a house that has not sold yet

70-467 - Machine Learning for Business Analytics

Andrés Castaño Zuluaga

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**Background.** The Ames City Assessor's Office used residential sales from 2006 to 2010 to keep its valuation model current. Ames is Iowa State's town. Those years run through the housing boom and the crash. When a house sold, the office recorded living area, overall quality, year built, baths, garage, kitchen, and neighborhood. After the closing it also recorded how the sale closed and whether the house was new. Partial in this file often means the house was not finished when it was last assessed.

You have 2,925 of those sales, in dollars. Five sales with more than 4,000 square feet of living area are already out. Mean price is about \$180,000. Partial and New sales sit around \$275,000; the rest sit around \$172,000. The codebook on the next pages gives type and unit for every column.

A common rule is living area. In this file overall quality moves tens of thousands of dollars; leaving it out will change the posted value.

The office still has to put a number on a house that is not in the file. Residual on the sales you already used is not residual on that house. Twenty-eight neighborhoods can look good on a training sample and worse on a test sample. A second training sample can tell a different story about those 28 names. Partial and New came in high here. Those labels exist only after the deal is done. They are not facts the office would have before a sale closes.

**The problem.** Produce a value for a house that is not in the file, using only facts the office would have before the sale closes, and report test MSE next to the \$180,000 mean. Please say whether neighborhood dummies belong, and whether Partial and New belong. One training sample is not enough to lock that choice. Average test MSE across several training samples, then lock a model.

Bring a laptop. After class you will have additional questions connected to the evaluation slides.

**Codebook.** 2,925 rows, 82 columns. One row = one residential sale in Ames, 2006–2010. Cite De Cock (2011), *Journal of Statistics Education*.

| Variable               | Type    | Unit  | Description                                                           |
|------------------------|---------|-------|-----------------------------------------------------------------------|
| <b>Order</b>           | numeric | none  | Row index in the original file.                                       |
| <b>PID</b>             | numeric | none  | Parcel ID. Not a predictor.                                           |
| <b>MS.SubClass</b>     | numeric | code  | Building class (one-story, two-story, ...).                           |
| <b>MS.Zoning</b>       | factor  | none  | Zoning: RL, RM, ...                                                   |
| <b>Lot.Frontage</b>    | numeric | feet  | Street frontage. Missing on 490 rows.                                 |
| <b>Lot.Area</b>        | numeric | sq ft | Lot size.                                                             |
| <b>Street</b>          | factor  | none  | Paved or gravel.                                                      |
| <b>Alley</b>           | factor  | none  | Alley access. Missing = no alley (most rows).                         |
| <b>Lot.Shape</b>       | factor  | none  | Lot shape.                                                            |
| <b>Land.Contour</b>    | factor  | none  | Flatness.                                                             |
| <b>Utilities</b>       | factor  | none  | Utilities available.                                                  |
| <b>Lot.Config</b>      | factor  | none  | Inside, corner, cul-de-sac, ...                                       |
| <b>Land.Slope</b>      | factor  | none  | Slope.                                                                |
| <b>Neighborhood</b>    | factor  | none  | 28 neighborhoods in Ames.                                             |
| <b>Condition.1</b>     | factor  | none  | Proximity to main road or railroad.                                   |
| <b>Condition.2</b>     | factor  | none  | Second proximity condition if any.                                    |
| <b>Bldg.Type</b>       | factor  | none  | One-family, townhouse, duplex, ...                                    |
| <b>House.Style</b>     | factor  | none  | One story, two story, ...                                             |
| <b>Overall.Qual</b>    | numeric | 1–10  | Overall material and finish. Large in dollars in this file.           |
| <b>Overall.Cond</b>    | numeric | 1–10  | Overall condition.                                                    |
| <b>Year.Built</b>      | numeric | year  | Year built.                                                           |
| <b>Year.Remod.Add</b>  | numeric | year  | Remodel year (same as built if none).                                 |
| <b>Roof.Style</b>      | factor  | none  | Roof style.                                                           |
| <b>Roof.Matl</b>       | factor  | none  | Roof material.                                                        |
| <b>Exterior.1st</b>    | factor  | none  | Exterior covering.                                                    |
| <b>Exterior.2nd</b>    | factor  | none  | Second exterior if any.                                               |
| <b>Mas.Vnr.Type</b>    | factor  | none  | Masonry veneer type.                                                  |
| <b>Mas.Vnr.Area</b>    | numeric | sq ft | Masonry veneer area.                                                  |
| <b>Exter.Qual</b>      | factor  | none  | Exterior quality.                                                     |
| <b>Exter.Cond</b>      | factor  | none  | Exterior condition.                                                   |
| <b>Foundation</b>      | factor  | none  | Foundation type.                                                      |
| <b>Bsmt.Qual</b>       | factor  | none  | Basement height/quality. Missing = no basement.                       |
| <b>Bsmt.Cond</b>       | factor  | none  | Basement condition.                                                   |
| <b>Bsmt.Exposure</b>   | factor  | none  | Walkout or garden exposure.                                           |
| <b>BsmtFin.Type.1</b>  | factor  | none  | Rating of finished basement area 1.                                   |
| <b>BsmtFin.SF.1</b>    | numeric | sq ft | Type 1 finished sq ft.                                                |
| <b>BsmtFin.Type.2</b>  | factor  | none  | Rating of finished basement area 2.                                   |
| <b>BsmtFin.SF.2</b>    | numeric | sq ft | Type 2 finished sq ft.                                                |
| <b>Bsmt.Unf.SF</b>     | numeric | sq ft | Unfinished basement sq ft.                                            |
| <b>Total.Bsmt.SF</b>   | numeric | sq ft | Total basement sq ft.                                                 |
| <b>Heating</b>         | factor  | none  | Heating type.                                                         |
| <b>Heating.QC</b>      | factor  | none  | Heating quality.                                                      |
| <b>Central.Air</b>     | factor  | none  | Central air Y/N.                                                      |
| <b>Electrical</b>      | factor  | none  | Electrical system.                                                    |
| <b>X1st.Flr.SF</b>     | numeric | sq ft | First floor sq ft.                                                    |
| <b>X2nd.Flr.SF</b>     | numeric | sq ft | Second floor sq ft.                                                   |
| <b>Low.Qual.Fin.SF</b> | numeric | sq ft | Low-quality finished sq ft.                                           |
| <b>Gr.Liv.Area</b>     | numeric | sq ft | Above-grade living area. Five sales over 4,000 sq ft already dropped. |
| <b>Bsmt.Full.Bath</b>  | numeric | count | Basement full baths.                                                  |
| <b>Bsmt.Half.Bath</b>  | numeric | count | Basement half baths.                                                  |
| <b>Full.Bath</b>       | numeric | count | Full baths above grade.                                               |
| <b>Half.Bath</b>       | numeric | count | Half baths above grade.                                               |
| <b>Bedroom.AbvGr</b>   | numeric | count | Bedrooms above grade.                                                 |
| <b>Kitchen.AbvGr</b>   | numeric | count | Kitchens above grade.                                                 |
| <b>Kitchen.Qual</b>    | factor  | none  | Kitchen quality.                                                      |
| <b>TotRms.AbvGrd</b>   | numeric | count | Rooms above grade (not baths).                                        |
| <b>Functional</b>      | factor  | none  | Home functionality.                                                   |
| <b>Fireplaces</b>      | numeric | count | Number of fireplaces.                                                 |
| <b>Fireplace.Qu</b>    | factor  | none  | Fireplace quality. Missing = no fireplace.                            |
| <b>Garage.Type</b>     | factor  | none  | Garage location. Missing = no garage.                                 |
| <b>Garage.Yr.Blt</b>   | numeric | year  | Year garage built.                                                    |
| <b>Garage.Finish</b>   | factor  | none  | Interior finish of garage.                                            |
| <b>Garage.Cars</b>     | numeric | count | Car capacity. One missing row in the raw file.                        |
| <b>Garage.Area</b>     | numeric | sq ft | Garage area.                                                          |
| <b>Garage.Qual</b>     | factor  | none  | Garage quality.                                                       |
| <b>Garage.Cond</b>     | factor  | none  | Garage condition.                                                     |
| <b>Paved.Drive</b>     | factor  | none  | Paved driveway.                                                       |
| <b>Wood.Deck.SF</b>    | numeric | sq ft | Wood deck.                                                            |

| Variable              | Type    | Unit  | Description                                                                   |
|-----------------------|---------|-------|-------------------------------------------------------------------------------|
| <b>Open.Porch.SF</b>  | numeric | sq ft | Open porch.                                                                   |
| <b>Enclosed.Porch</b> | numeric | sq ft | Enclosed porch.                                                               |
| <b>X3Ssn.Porch</b>    | numeric | sq ft | Three-season porch.                                                           |
| <b>Screen.Porch</b>   | numeric | sq ft | Screen porch.                                                                 |
| <b>Pool.Area</b>      | numeric | sq ft | Pool area (almost all zero).                                                  |
| <b>Pool.QC</b>        | factor  | none  | Pool quality. Missing on every remaining row.                                 |
| <b>Fence</b>          | factor  | none  | Fence quality. Missing = no fence.                                            |
| <b>Misc.Feature</b>   | factor  | none  | Shed, etc. Missing = none.                                                    |
| <b>Misc.Val</b>       | numeric | USD   | Value of miscellaneous feature.                                               |
| <b>Mo.Sold</b>        | numeric | month | Month the sale closed.                                                        |
| <b>Yr.Sold</b>        | numeric | year  | Year the sale closed (2006–2010).                                             |
| <b>Sale.Type</b>      | factor  | none  | How it sold (WD, New, COD, ...). Recorded at closing.                         |
| <b>Sale.Condition</b> | factor  | none  | Normal, Partial, Abnorml, Family, ...Recorded at closing. Partial $n = 242$ . |
| <b>SalePrice</b>      | numeric | USD   | Sale price. This is $Y$ . Mean about \$180,412 after the living-area screen.  |